Representation Learning with Autoencoders

AE & β-VAE on Medical MNIST

DSAI 490 – Assignment 1

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Problem and dataset

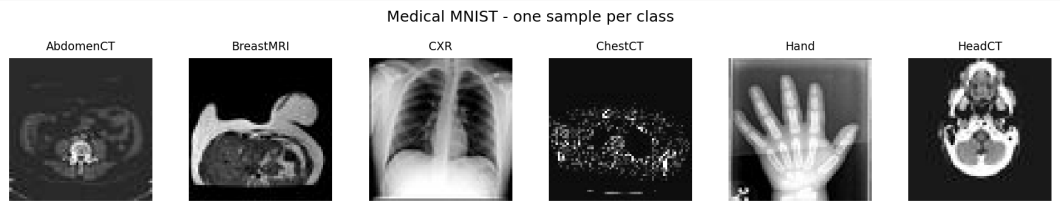
Goal. Compare a convolutional Autoencoder and a convolutional β-Variational Autoencoder on real medical images, measuring reconstruction, latent structure, generation, and denoising.

Dataset. Medical MNIST – 6 classes of 64×64 grayscale scans: AbdomenCT, BreastMRI, CXR, ChestCT, Hand, HeadCT.

Pipeline. `tf.data` – JPEG decode $\rightarrow$ resize $\rightarrow$ $[0, 1]$ scale $\rightarrow$ shuffle $\rightarrow$ batch 128 $\rightarrow$ prefetch. Split 80/10/10 train/val/test.

Training. Keras functional API, Adam (10^{-3}), **30 epochs**, single Kaggle GPU.

Sample images and classes



Six visually distinct modalities and anatomies. CXR and ChestCT are most similar to each other; the others are well-separated.

Architectures – AE vs β-VAE

Autoencoder

- `Conv2D` + `MaxPool` $\times 3$ ($32 \to 16 \to 16$)
- Bottleneck: $8 \times 8 \times 16 = 1024$ dims
- `Conv2DTranspose` $\times 3$, stride 2
- Sigmoid output head
- Loss: pixelwise BCE
- Deterministic, unregularised

β-VAE

- Strided `Conv2D` ($32 \to 64 \to 64$) + dense 128
- Two heads $\to (\boldsymbol{\mu}, \log \boldsymbol{\sigma}^2) \in \mathbb{R}^{16}$
- Reparameterise $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}$
- Symmetric `Conv2DTranspose` decoder
- Loss: BCE $+ \beta \cdot \mathrm{KL}(q \,\|\, \mathcal{N}(0, I))$
- Probabilistic, prior-regularised

Why $\beta = 0.25$ and not $\beta = 1$?

The standard ELBO uses $\beta = 1$. On this dataset, that caused *partial posterior collapse*:

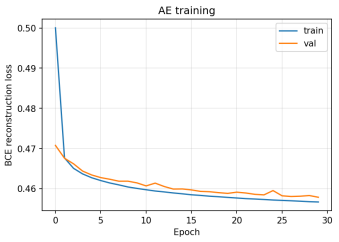
- KL dropped to ≈ 27 nats (encoder hugging the prior)
- Reconstructions were severely blurred
- The encoder gave up information to minimise KL

Lowering β to 0.25 lets the encoder carry more information:

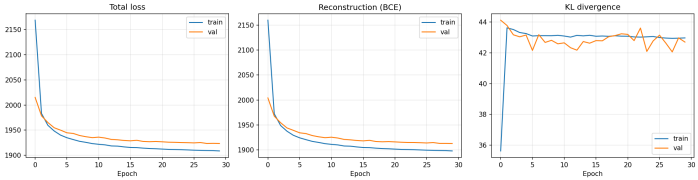
- KL rose to ≈ 43 nats (counter-intuitive but correct: lower $\beta \Rightarrow$ *higher* KL at equilibrium)
- Reconstruction BCE dropped from ~ 1970 to ~ 1900
- Generation and denoising both improved

This is β-VAE (Higgins et al., 2017), with $\beta < 1$ favouring reconstruction.

Training dynamics

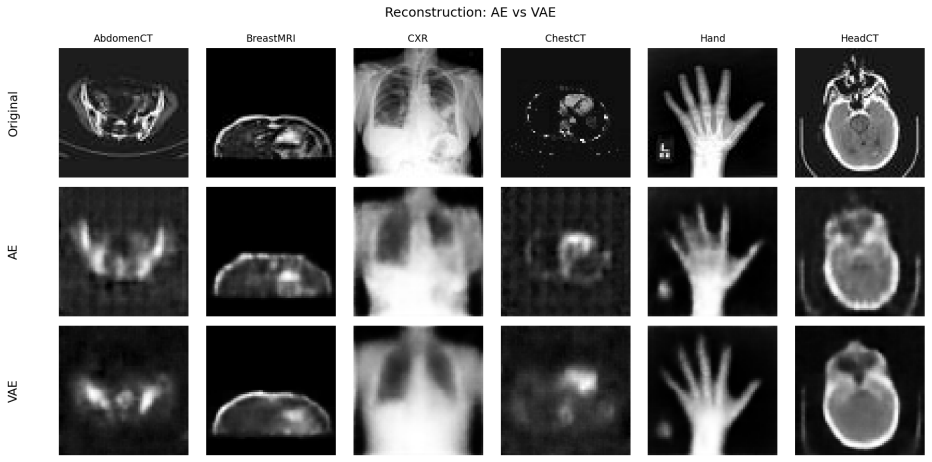


AE. BCE 0.50 → 0.46, plateau after ∼20 epochs.



β-**VAE.** Reconstruction drops 2160 → 1900; KL settles at ∼43 nats. Encoder carrying information, not collapsing.

Reconstruction: AE vs β-VAE



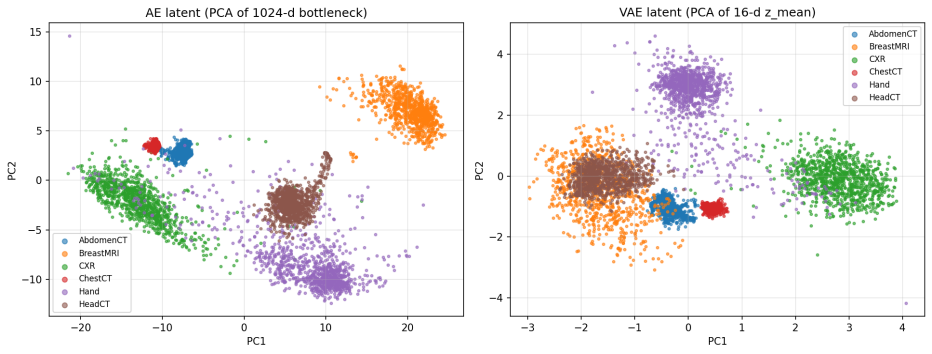
Both preserve global anatomy. AE preserves slightly more texture (1024-dim code). β-VAE output is smoother – the "VAE blur," driven by the 16-dim KL-regularised code.

Reconstruction metrics

	Test MSE	Test MAE
AE	0.00429	0.0355
β-VAE	0.00699	0.0449

- AE wins on pixel error by $\approx 39\%$ – expected: 1024 free dimensions, no KL regulariser.
- VAE "loses" by design: 16 regularised dimensions + explicit pull toward the prior.
- Low numbers are misleading on their own – MSE is dominated by the large black backgrounds.

Latent-space structure



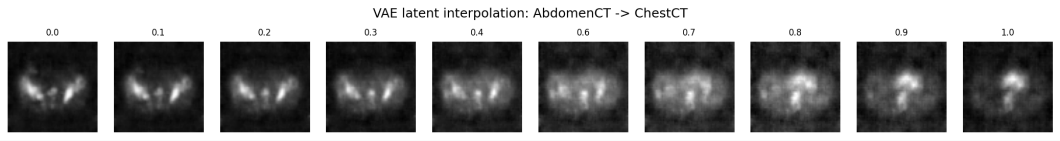
AE (left). Tight, well-separated class clusters – good for a downstream classifier. β-**VAE (right).** Compact, origin-centered, more overlap – the KL pull, but densely populated where the prior puts mass.

Sampling from the VAE prior

64 samples, $z \sim \mathcal{N}(0, I)$

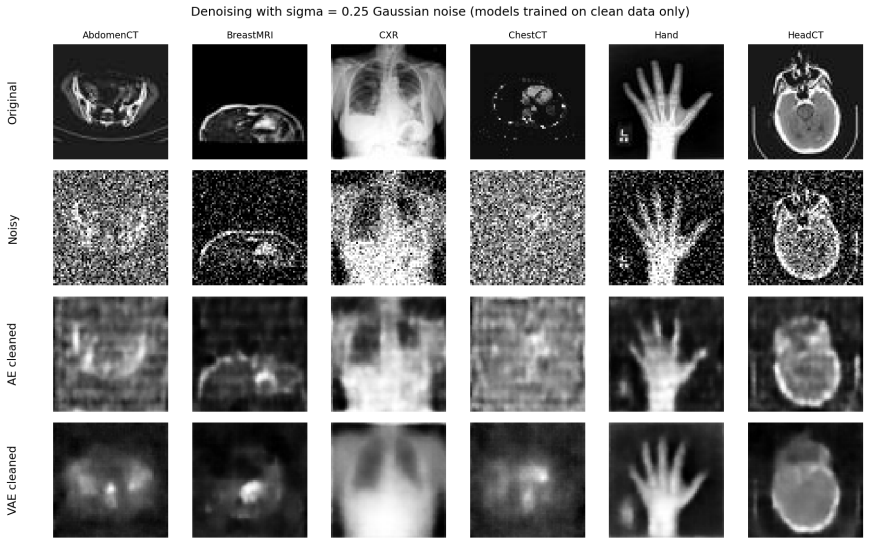
- Hands and chest silhouettes *recur* across the grid – real class structure learned in the prior.
- Samples are blurry (small latent + Gaussian likelihood, not a training issue).
- Sharper synthesis would require VQ-VAE or diffusion at much higher cost.

Latent interpolation



Linear interpolation in μ-space between an AbdomenCT and a ChestCT encoding. Smooth pixel-level transitions (no jumps) confirm the latent manifold is continuous – a prerequisite for semantic editing. Intermediate frames are visually ambiguous on this dataset.

Denoising – emergent, without retraining



Both models trained on clean data only; fed inputs with $\sigma = 0.25$ Gaussian noise

Key findings

1. **AE and β-VAE are not interchangeable.** They optimise different objectives.
2. **AE wins on reconstruction & classification.** 39% lower MSE, tighter class clusters in latent space.
3. **β-VAE wins on generation & denoising.** Only model that can sample; cleaner denoising thanks to the regularised latent.
4. **β-tuning matters.** Standard β=1 caused partial posterior collapse on this dataset; β=0.25 fixed it.
5. **Counter-intuitive fact.** Lowering β *raises* the equilibrium KL – the encoder is allowed to carry information.

Limitations and future directions

- **Small latent + Gaussian prior is a ceiling for sample sharpness.** VQ-VAE, hierarchical VAEs, or diffusion models would sharpen generation.
- **Denoising was emergent, not trained.** A noise-conditioned denoising AE (noisy input, clean target) would outperform both models for this specific task.
- **Medical MNIST is easy.** Classes are visually distinct, sizes are small. Conclusions do not directly transfer to full-resolution clinical imaging.
- **No perceptual loss.** Pixel BCE underweights structural detail. Using a perceptual loss (e.g. from a pretrained classifier) would likely improve both AE and VAE reconstructions.